

Initializing: On power up the system runs startup checks and initializes the sensors, logging, telemetry and recovery systems.

Fault: This state is only entered if a problem is detected. If the fault is cleared it returns to the previous state. If the fault cannot be cleared the system will restart back to initialization state essentially rebooting the system.

Wait For Launch: After Initialization, the system is armed and waiting for launch detection. It monitors sensors, records data and checks system health while transmitting telemetry

Ascent: Once launch has been detected the system enters the ascent phase, it records data and monitor sensors data at a high rate to accurately detect apogee with minimal error using multi sensor fusion for detection.

Descent: Once the system detects apogee where the rocket starts falling back towards the Earth, the recovery sequence begins.

Drogue Descent: A fire drogue command is sent out to deploy the drogue chute; this small chute lets the rocket to descent as quickly but in a stable and controlled manner. The system continues to monitor altitude and waits for the main deployment point.

Main Descent: A fire main command is sent out to deploy the main chute; this large chute brings the rocket descent down dramatically so it can ensure a slow and safe landing speed.

Landed: Once no movement is detected for a set period of time the system enters the landed state. It will disable non essential systems to conserve power while transmitting its GPS location to aid in recovery.

